Antonio Rafael Vazquez Pantoja

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Education

Texas A&M University May 2028
Bachelor of Science in Mechanical Engineering, Minor in Biomedical Engineering *GPA: 3.4*

Experience

Biotex Inc. May 2026 - Aug 2026
R&D Mechanical Engineering Intern *Houston, TX*

- Built and iterated a motorized puncture test fixture to automate needle rotation, puncture motion, and reset positioning, enabling 20,000 continuous test cycles where FDA regulations limit a single needle to 50 manual punctures.
- Designed modular fixture components using screws, heat-set inserts, and 3D-printed parts to support rapid assembly and iterative prototyping.
- Improved manufacturability and reduced fracture risk in fixture components by applying DFM principles including wall-thickness optimization, strain analysis, and redesign of components for CNC machining and prototype fabrication.
- Maintained compliant engineering documentation within a regulated medical-device R&D environment, as required by GMP and ISO 13485 quality system practices, by documenting prototype testing, design updates, and engineering changes throughout development.

Texas A&M – Biomedical Engineering Department Jan 2026 - Present
Undergraduate Research Assistant, Ethicon (Johnson and Johnson Sponsored Research) *College Station, TX*

- Conduct corrosion-mechanical coupled testing on bioabsorbable magnesium surgical staples to characterize degradation-driven changes in structural integrity and mechanical performance.
- Determine minimum and maximum bend radius constraints through tensile stress–strain testing, identifying elastic limits, yield behavior, and failure points across varied bend geometries.
- Analyze how progressive corrosion influences mechanical response, supporting evaluation of geometric and material failure modes relevant to bioresorbable implant design.

Texas A&M – Biomedical Engineering Department Sept 2025 - Jan 2026
Undergraduate Research Assistant, Cardiovascular Devices *College Station, TX*

- Assembled multiple RF balloon catheter prototypes for fetal hypoplastic left heart syndrome (HLHS), soldering RF PCB components to the balloon outer layer for bench-top and porcine tissue-based evaluation of RF energy delivery.
- Iterated through three test bed designs to assess RF-induced tissue denaturing in aqueous conditions; resolved tissue tearing and inconsistent contact by transitioning from a rectangular to circular geometry to distribute load evenly across the sample.
- Incorporated silicone fixturing in the final design iteration to improve tissue positioning and PCB contact, increasing test repeatability and RF energy transfer consistency.

Texas A&M - Mechanical Engineering Department June 2025 - Present
Undergraduate ME Engineering Research Assistant *College Station, TX*

- Co-authored peer-reviewed publication on the ANGEL soft robotic gripper; designed CAD models for primary structural components, motor spool geometry, and cable routing architecture to optimize wire tension and contact force distribution.
- Designed and iterated 3D-printed TPU pocket geometries (cushioning, flexure, and contact area variants) to minimize surface damage during grasping.
- Conducted experimental trials across 56 tomatoes, achieving a 0% immediate damage rate and under 9% bruise rate at 5 days post-harvest.

Del Sol Foods Co. Oct 2023 - Present
Manufacturing and Industrial Engineering Intern *Brenham, TX*

- Collaborated with Production, Maintenance, and Quality teams to troubleshoot equipment failures, reduce unplanned downtime, and improve throughput in a high-volume manufacturing environment.
- Developed and maintained SOPs and manufacturing documentation to support quality and compliance.

Projects

Turtle Robotics – LARM: Laboratory Robotic Manipulator Sept 2025 - May 2026

- Leading end-to-end mechanical design of a servo-actuated gripper for manipulation of 500 mL laboratory beakers in wet lab environments.
- Engineered a 3-gear transmission system to synchronize claw motion and amplify torque from a 20 kg·cm servo motor.
- Evaluated slip risk on wet glass surfaces and integrated silicone contact pads to increase friction coefficient; designed claw curvature to conform to cylindrical glassware geometry for improved load distribution.
- Iterated CAD models and tolerance stack-ups in SolidWorks to ensure smooth gear meshing and repeatable actuation.

Technical Skills

CAD % Design: Inventor, SolidWorks (Parts, Assemblies, Drawings), OnShape, AutoCAD, Tolerance Stack-Up, DFM
Testing & Analysis: Corrosion-mechanical coupled testing, tensile testing, stress–strain analysis, bend radius characterization, failure mode characterization, accelerated corrosion testing
Programming & Simulation Python, FluidSIM, NI Multisim, LaTeX.
Manufacturing & Hardware: Rapid Prototyping, 3D Printing (PLA, PETG, TPU), Design for Manufacturability (DFM), Mechatronics, PCB Design, PLC Programming, Soldering
Languages: English (Fluent), Spanish (Fluent)

Publications

Patel, D., **Vazquez Pantoja, A. R.**, Lei, J., Lee, K., Liang, X., & Zheng, M. (2025). *ANGEL: A novel gripper for versatile and light-touch fruit harvesting.*